

Decision Making Under Uncertainty

Exam Cheat Sheet — Linear Programming, Sensitivity Analysis & Network Flow

1 Linear Programming (LP) Formulation

An LP is just a clean, math way of writing down a decision problem: what you're choosing, what's stopping you, and what you're trying to achieve.

The 4 pieces — always write them in this order

- **1. Parameters** — the numbers you were given (costs, capacities, requirements). Just restate them, often as a small table.
- **2. Decision Variables** — what you get to choose. Give each one a letter and a plain description.
- **3. Constraints** — the limits/rules. One line per rule, written as: (short description): (the math).
- **4. Objective Function** — the one thing you Maximize or Minimize.

Example: You run a bakery. You bake cookies (x_1) and muffins (x_2). Decision variables: x_1 = number of cookies, x_2 = number of muffins. If flour is limited to 100 cups and each cookie needs 1 cup, each muffin needs 2 cups: constraint is $x_1 + 2x_2 \leq 100$.

Common constraint "shapes" — recognize these instantly

What it's saying (plain English)	How to write it
You can't use more of a resource than you have	$(\text{usage per unit}) \cdot x_1 + (\text{usage per unit}) \cdot x_2 \leq \text{available}$
There's a cap on how many of something you can make/use	$x_1 \leq 10$
You must make/serve at least this many	$x_1 + x_2 \geq 60$ or $x_1 \geq 10$
One thing can't be more than a % of a total	$600x_1 \leq 0.7 \cdot (600x_1 + 470x_2)$
One thing must be at least a multiple of another	$x_1 \geq 2 \cdot x_3$
You can never choose a negative amount	$x_1, x_2 \geq 0$ (never forget this line!)

If asked to explain a constraint in words

Say what's on the LEFT (what you're using up or producing) versus what's on the RIGHT (the limit), **in the context of the real problem** — don't just translate symbols back into English word-for-word.

2 Sensitivity Analysis (reading the Excel report)

This is the biggest chunk of the exam. You're given a Solver Sensitivity Report (two tables) and asked questions **without re-solving the problem**.

The two tables — what each column means, in plain English

Column	Plain-English meaning
Final Value (variable)	The actual optimal amount of that variable — this IS your answer.
Reduced Cost	How much better that variable's "payoff" would need to be before it's worth using it at all (usually 0 if it's already being used).

Objective Coefficient	The cost/profit per unit you were given for that variable.
Allow. Increase / Decrease (variable)	How much you can change that coefficient before the OPTIMAL MIX (which variables, how much) might change.
Final Value (constraint)	How much of that resource you actually used.
Shadow Price	The change in the objective value for ONE more unit of that resource. 0 if there's slack (not binding).
Constraint R.H. Side	The original limit you were given.
Allow. Increase / Decrease (constraint)	The range the limit can move within before the shadow price stops being valid.

"Binding" — the most important word on this exam

A constraint is BINDING when: Final Value = Constraint R.H. Side

In plain English: you're using every last bit of that resource (or hitting the minimum exactly). No wiggle room left. If Final Value is LESS than the R.H. Side, there's leftover ("slack") and the shadow price is automatically 0.

Answer templates — copy these patterns on exam day

Q: "How many of X should we make/build?"

A: Read the Final Value straight off the Variable Cells table. State it in words: "We should build 3 new parks."

Q: "Did we use all of resource Y? How much is left?"

A: Compare Final Value vs. R.H. Side for that constraint. If equal -> fully used, nothing left. If Final Value is smaller, the difference is the leftover amount.

Q: "Interpret the shadow price of constraint Z"

A: "If we had one more unit of [resource], the objective would improve by [shadow price]. This only holds as long as the resource stays between [R.H.Side - Allow. Decrease] and [R.H.Side + Allow. Increase]. Outside that range we'd have to re-solve." — ALWAYS include the range, it's graded separately.

Q: "If a coefficient changes (e.g., a new survey estimate), does the recommendation change?"

A: Check if the new value is still inside [current - Allow. Decrease, current + Allow. Increase] for that variable. Inside the range -> same optimal mix, recommendation doesn't change. Outside the range -> can't say for sure without re-solving.

Q: "How should we use the remaining budget on resource that costs \$c/unit?"

A: Compare the shadow price of that (binding) resource to its cost \$c. Shadow price > cost -> worth buying more (keeps helping) until you hit the Allowable Increase limit. Shadow price <= cost -> not worth it.

Q: "Are multiple optimal solutions possible?"

A: Yes if the Allowable Increase or Decrease of an objective coefficient is exactly 0 for some variable — that's the tell-tale sign.

! GOLDEN RULE: Sensitivity analysis only lets you change ONE thing at a time. Never add two shadow prices together, or combine a coefficient change with a resource change — that's a classic trap answer marked False.

Example: Shadow price of packing-time = \$10, Allowable Increase = 10, Allowable Decrease = 70, R.H. Side = 70 hours. Meaning: for every extra hour of packing time (up to 70+10=80 hours total), profit goes up \$10. Below 70-70=0 hours the interpretation stops applying too.

3 Network Flow Models

Used when you're moving/assigning things between locations, people, or categories (shipping, staffing, assignments). Think of it as dots (nodes) connected by arrows (arcs).

The 3 types of nodes

Node type	What it means	How to write its constraint
Supply / Origin	A place that HAS a fixed amount to send out	sum of outgoing arcs = supply (e.g. $x_{1,4} + x_{1,5} = 200$)
Demand / Destination	A place that NEEDS a fixed amount to arrive	sum of incoming arcs = demand (e.g. $x_{4,6} + x_{5,6} = 150$)
Intermediate / Transshipment	A pass-through point (supply = demand = 0)	inflow = outflow (e.g. $x_{1,4} + x_{2,4} + x_{3,4} = x_{4,6} + x_{4,7} + x_{4,8}$)

Full template to write on your exam

- **Decision variables:** x_{ij} = amount/flow (or 1-if-assigned) from node i to node j
- **Flow conservation constraints:** one per node, using the table above
- **Capacity constraints** (if arcs have limits): $x_{ij} \leq \text{capacity of that arc}$
- **Non-negativity:** $x_{ij} \geq 0$ for every arc
- **Objective:** Maximize or Minimize the sum of (cost or benefit) $\cdot x_{ij}$ over every arc

Drawing the diagram

Circles = nodes (label with supply/demand number). Arrows = arcs (label with capacity and cost/benefit). Layout: origins on the left -> intermediates in the middle -> destinations on the right.

Special case: Assignment problems (e.g., events->parks, mentors->mentees)

Same idea, just: supply side often has a max capacity per node ("each park hosts ≤ 2 events" -> use $\leq$), demand side usually needs exactly 1 match ("every event goes to exactly one park" -> use $=$).

Special case: Two things flowing together (multicommodity)

If two different resources move through the same arcs (e.g. pilots AND copilots on the same flights), use two separate variables (x_{ij} , y_{ij}) with their own conservation constraints, then add one linking constraint tying them together, e.g. $x_{ij} + y_{ij} = 2 \cdot (\text{flights on that arc})$, since each flight needs exactly 2 crew.

4 Quick Glossary (no jargon)

Term	Means
Decision variable	The thing you're choosing the amount of.
Objective function	The one number (profit, cost, benefit) you're trying to make as big or small as possible.
Feasible region	All the combinations of your variables that don't break any rule.
Binding constraint	A rule you're using up completely — no room left.
Non-binding / slack	A rule you have leftover room on — it's not the thing holding you back.
Shadow price	How much better/worse your result gets for one more/less unit of a binding resource.
Allowable increase/decrease	How far a number can move before your answer might need to change.

Unbounded	The objective can improve forever — usually means a constraint is missing.
Infeasible	There's no way to satisfy every rule at once — no valid solution exists.
Multiple optimal solutions	More than one combination gives the exact same best result.

80-minute exam pacing: ~20-25 min LP formulation, ~30-35 min sensitivity analysis, ~20-25 min network flow. Always finish with units/context in your final answer, and never forget the non-negativity line.